

Cold-Plate Design Challenge: Vikings Group Design

Amirhossein Bayat

*Institute of Mechanical and Electrical Engineering
University of Southern Denmark
Odense, Denmark
abay@sdu.dk*

Magnus Appel

*Institute of Mechanical and Electrical Engineering
University of Southern Denmark
Odense, Denmark
magap@sdu.dk*

Hao Li (Supervisor)

*Institute of Mechanical and Electrical Engineering
University of Southern Denmark
Odense, Denmark
hli@sdu.dk*

Abstract—This paper presents the methodology and results of the Cold Plate Design Challenge. We developed a hybrid heat sink that combines pin fins and bar fins, where the bar-fin placement is guided by topology optimization, enabling effective use of additive manufacturing due to its high geometric freedom. This approach leverages the heat-transfer advantages of both fin types while substantially reducing the pressure drop, thereby maintaining low temperatures and enhancing the overall Figure Of Merit (FOM). The resulting design achieved an FOM of 0.045, indicating improved performance compared with the provided baseline design. For the topology optimization, we employed a scaling strategy and a reduced-dimensional, two-layer model to retain as much physical fidelity as possible within a computationally efficient framework. Subsequently, additional geometric modifications were introduced to further improve performance relative to the baseline design, including shear features to reduce the pressure drop and the addition and optimization of pin fins to maximize the design FOM. Overall, this work highlights the potential of the proposed topology-optimization approach for practical thermal management applications. The designed cold plate strictly adheres to the specified aspect-ratio, overhang-angle, and minimum feature-size constraints, ensuring compatibility with additive manufacturing.

A. Paper layout

In Section -B, we present the governing equations used in this study, including the fluid-flow equations, the reduced-dimensional heat-transfer model, and the topology-optimization formulation. In Section -C, we describe the design procedure and summarize the steps taken to obtain the final optimized geometry. The simulation results are discussed in Section -D.

B. Governing physics

1) *Incompressible laminar fluid flow*: A two-dimensional, steady-state, non-dimensional, and incompressible laminar flow is considered. The balance of linear momentum in non-dimensional form, in a two-dimensional domain, reduces to the following equation, known as the non-dimensional Navier–Stokes equations:

$$(\mathbf{u} \cdot \nabla)\mathbf{u} + \nabla p - \frac{1}{\text{Re}} \nabla \cdot (\nabla \mathbf{u} + \nabla \mathbf{u}^\top) - \mathbf{f} = 0 \quad (1)$$

where Re is the Reynolds number defined as:

$$\text{Re} = \frac{\rho U L}{\mu} \quad (2)$$

A volumetric force term is introduced to the Navier–Stokes equations to represent the effect of solid material on the fluid flow, formulated through the Brinkman penalty term [1]:

$$\mathbf{f} = \alpha \mathbf{u} \quad (3)$$

Here, α is the inverse permeability, which depends on the design variable γ , and it is defined as:

$$\alpha = \alpha_{\max} \frac{1 - \gamma}{1 + q_\alpha \gamma} \quad (4)$$

Incompressibility of the fluid is accounted for by the continuity equation, as follows:

$$\nabla \cdot \mathbf{u} = 0 \quad (5)$$

which ensures mass conservation within the domain. An example of the boundary conditions applied in the Navier–Stokes equations is where a parabolic velocity profile is prescribed at the inlet, a zero-pressure condition with a straight-out velocity is imposed at the outlet, and no-slip or free-slip boundary conditions are applied on all other outer boundaries.

2) *Governing Heat-Transfer Equations for Topology Optimization*: The design-optimization heat transfer model is inspired by the multi-layer approach proposed by Yan et al. [2]. They introduced a computationally efficient topology-optimization framework for microchannel heat-sink design by reducing the original three-dimensional problem to a tractable two-layer representation (Fig. 1). This representation consists of a modified advection–diffusion equation and a modified diffusion equation. Consequently, the governing equations can be written as:

$$\frac{3}{2} \rho C (\mathbf{u} \cdot \nabla T_t) - \frac{49}{52} k_t (\nabla^2 T_t) - \frac{h}{2H_t} (T_b - T_t) = 0 \quad (6)$$

$$-\frac{k_b}{2} \nabla^2 T_b + \frac{h}{2H_b} (T_b - T_t) - \frac{q_0''}{2H_b} = 0 \quad (7)$$

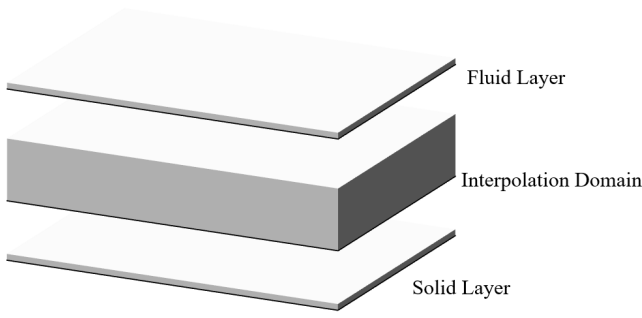


Fig. 1: Illustration of the layered structure used in the design framework. The fluid layer (top) represents the fluid domain, while the solid layer (bottom) represents the base domain. The interpolation domain (middle) facilitates the coupling between the fluid and solid layers, enabling the integration of multiphysics simulations in topology optimization.

where C is the heat capacity of the fluid. The fluid bulk-mean temperature, T_t , and the three-dimensional temperature field of the substrate, T_b , are the state variables in this reduced two-dimensional governing model. The parameters k_f and k_s denote the thermal conductivities of the fluid and solid, respectively. The term q_0'' is the out-of-plane applied heat flux. The parameter h is an effective interlayer convection coefficient that represents the coupling between the layers, while h_b and h_t are the out-of-plane heat transfer coefficients describing the heat transfer from the substrate to the thermal-fluid layer and vice versa.

In the topology optimization, an interpolation of the effective thermal conductivity, k_t , is introduced to distinguish between solid and fluid regions. In solid regions, k_t is set to its maximum value to represent the solid structure, whereas in fluid regions it is set equal to the fluid thermal conductivity. In addition, the out-of-plane heat transfer coefficient is also interpolated to better capture the distinct thermal characteristics of the solid and fluid phases. The Brinkman [3] interpolations are defined as:

$$k_t(\gamma) = k_f + (k_s - k_f) \frac{1 - \gamma}{1 + q_k \gamma} \quad (8)$$

$$h_t = \frac{35k_t}{2H_t} \quad (9)$$

$$h_b = \frac{k_b}{H_b} \quad (10)$$

$$h = \frac{h_t h_b}{h_t + h_b} \quad (11)$$

$$k_b = k_s \quad (12)$$

In this study, a velocity scaling strategy is adopted to address numerical challenges arising when the Reynolds number exceeds the solver's stable range, as proposed by Sun et al. [4]. The key idea is to scale the velocity field down

to a computationally feasible regime, solve the Navier-Stokes equations at this scaled level, and then rescale the velocity field back to its original magnitude. The process begins by selecting a scaled Reynolds number Re_s and calculating a scaling factor ζ as:

$$\zeta = \frac{Re_o}{Re_s} \quad (13)$$

The velocity and pressure fields are scaled using η and a pressure-specific scaling factor ζ_p :

$$\mathbf{u}_s = \frac{\mathbf{u}_o}{\zeta} \quad p_s = \frac{p_o}{\zeta_p} \quad (14)$$

Here, ζ_p , the pressure scaling factor, depends on the flow regime. For laminar flows, ζ_p is proportional to ζ , whereas for turbulent flows, it follows a quadratic relationship.

After solving the Navier-Stokes equations with the scaled fields and adjusted boundary conditions, the velocity field is rescaled to its original magnitude:

$$\mathbf{u}_o = \zeta \mathbf{u}_s \quad (15)$$

This rescaled velocity field is then used in the heat transfer equation to accurately model convective effects. By stabilizing the solver under high Reynolds number conditions, this strategy ensures accurate and efficient topology optimization.

3) *Objective function*: The objective function for the topology optimization problem is formulated as a multi-objective metric based on a modified Figure Of Merit (FOM). The goal is to maximize a weighted sum that combines the normalized improvements in thermal and hydraulic performance, where the pressure-drop term is scaled using the two-dimensional pressure coefficient η . In the thermal term, the hotspot temperature is replaced by the normalized N -norm temperature. The resulting objective function J is written as:

$$J = 0.7 \left(\frac{R_{th,ref} - R_{Nnorm}}{R_{th,ref}} \right) + 0.3 \left(\frac{\Delta P_{ref} - \eta \Delta P}{\Delta P_{ref}} \right) \quad (16)$$

where:

- η = 2D pressure scaling factor.
- $R_{Nnorm} = \frac{N_{norm} of T - T_{in}}{P_{load}}$ represents the thermal resistance, defined by the temperature difference between the hotspot and the inlet fluid over the load pressure.
- $\Delta P = P_{out} - P_{in}$ denotes the pressure drop across the heat sink.
- m is the mass of the fin plate, with m_{solid} as its reference solid mass.

This paper provides the practical implications of the proposed optimization framework for heat sink design, pushing the boundaries of thermal engineering to have a coupling between two- and three-dimensional topology optimization.

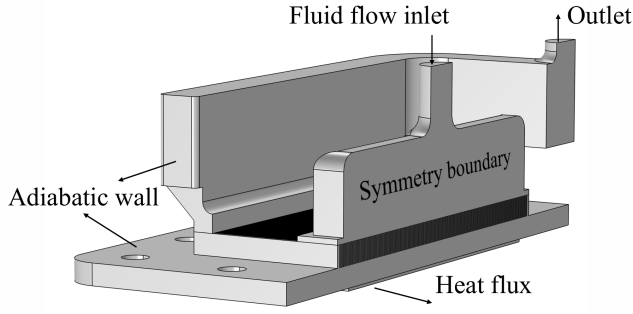


Fig. 2: Three-dimensional heat-sink geometry with symmetry boundary conditions.

4) *Optimization formulation:* The topology optimization problem is then formulated as

$$\begin{aligned} \max_{\gamma} \quad & J \\ \text{s.t.:} \quad & \text{Governing Eqs.} \\ & g_i(\gamma) \leq 0, \quad i = 1, \dots, m, \\ & 0 \leq \gamma \leq 1. \end{aligned}$$

where, γ_e is the design variable of element e and J is the objective function written in (16).

C. Design description

1) *Domain simplification:* The original model is fully three-dimensional, which leads to a prohibitively large meshing effort and a significant increase in the number of degrees of freedom (DOFs). To reduce the computational cost, the geometry is simplified in two steps. First, a symmetry plane is exploited (see Fig. 2), effectively halving the computational domain. Second, a computationally efficient two-dimensional equivalent model is constructed for the topology-optimization study (see Fig. 3).

2) *Topology optimization:* In the second step, topology optimization is employed to obtain a mathematically optimal design layout. The final optimized geometry is shown in Fig. 4. The optimization is performed in COMSOL Multiphysics 6.3 [5] using the Method of Moving Asymptotes (MMA) together with discrete-adjoint sensitivity analysis. In total, 400 iterations are carried out to reach the final design.

3) *Adding and tuning features:* Post-processing of topology optimized geometry was performed in AutoDesk Inventor [6] to further improve the figure Of merit (FOM) of the topology-optimized design. First, shear features were introduced to preserve the main topology-optimization (TO) features while significantly reducing the pressure drop. This modification was motivated by the observation that, although TO reduces flow penetration into solid regions, small residual flow paths may still exist; the added shear features help capture and reinforce these TO-informed flow effects in the final manufacturable geometry. Second, pin fins were added to enhance convective heat transfer and promote a more uniform temperature

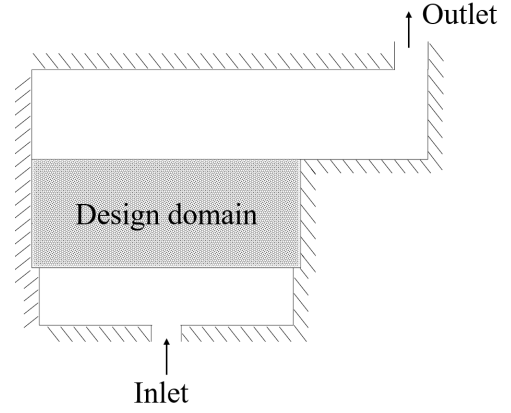


Fig. 3: Two-dimensional equivalent geometry of the heat sink. The design domain is the region used for topology optimization and represents the part intended to be manufactured using additive manufacturing.



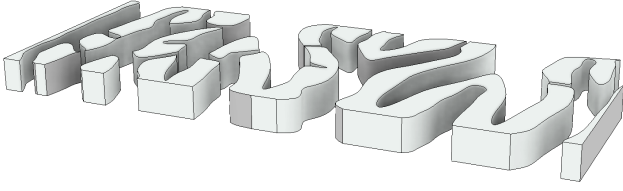
Fig. 4: Optimized 2D layout obtained from the reduced-dimensional topology-optimization model.

distribution, while maintaining the overall flow structure and avoiding excessive flow blockage. The pin-fin spacing was subsequently optimized by evaluating six candidate distances ranging from 600 μm to 2.1 mm (see Fig. 6). Among the tested configurations, a spacing of 900 μm yielded the highest FOM. The resulting optimized heat-sink geometry is shown in Fig. 7.

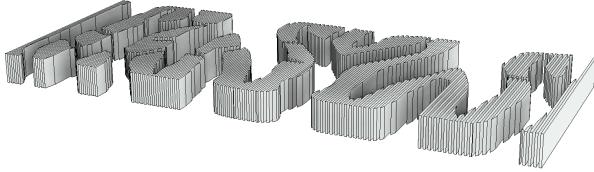
D. Simulation results

In this section, we present the numerical results for the final optimized geometry shown in Fig. 7, together with the corresponding Figure Of Merit (FOM) metrics reported in Table I. The temperature, pressure, and velocity fields for the final design are shown in Fig. 8. The velocity distribution remains relatively uniform throughout the domain, while the topology-optimized bar-fin features guide the flow and promote mixing. In addition, the pin fins visibly enhance convective heat transfer and contribute to a more homogeneous thermal field.

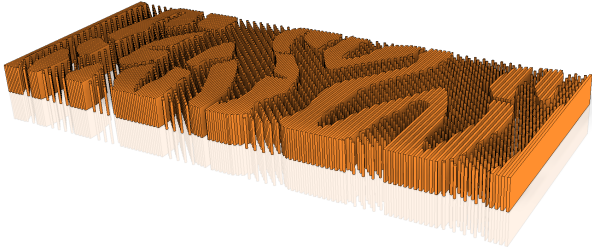
The temperature contours also indicate that the optimized layout mitigates thermal non-uniformities near the outlet: the longer flow path on the right-hand side effectively promotes lateral redistribution of the flow before exiting, which reduces



(a) Mapping the optimized 2D layout to the 3D cold-plate geometry.



(b) Introducing shear features to reduce the pressure drop while preserving consistency with the topology-optimized layout.



(c) Adding and tuning pin fins to enhance heat transfer with minimal impact on the flow field.

Fig. 5: Design evolution from the reduced-dimensional topology-optimization result to the final 3D heat-sink geometry: (a) 2D-to-3D mapping, (b) addition of shear features to reduce pressure losses, and (c) incorporation of pin fins to improve heat transfer while limiting additional hydraulic resistance.

peak temperatures and improves temperature uniformity. As summarized in Table I, the maximum temperature is 72.5°C and the average inlet-to-outlet pressure drop is 1200.3 Pa , resulting in an overall FOM of 0.045 .

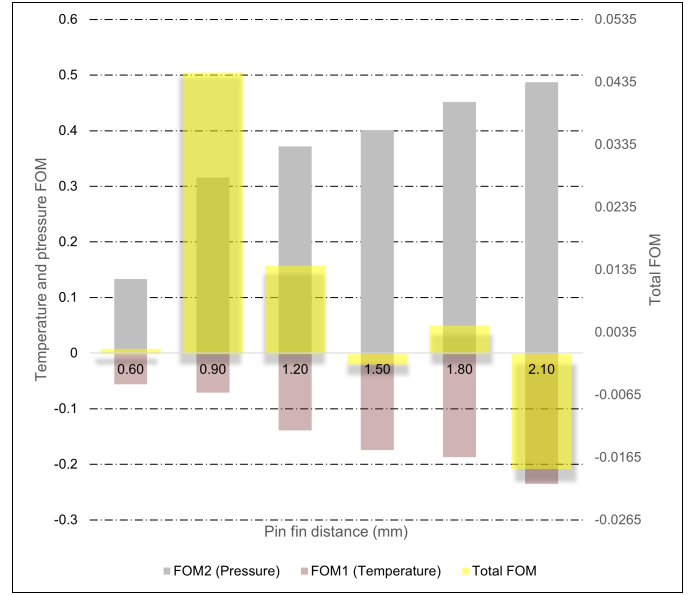


Fig. 6: Illustration of the layered structure used in the design framework. The fluid layer (top) represents the fluid domain, while the solid layer (bottom) represents the base domain. The interpolation domain (middle) facilitates the coupling between the fluid and solid layers, enabling the integration of multiphysics simulations in topology optimization.

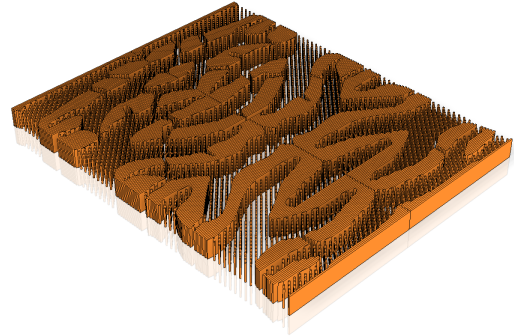
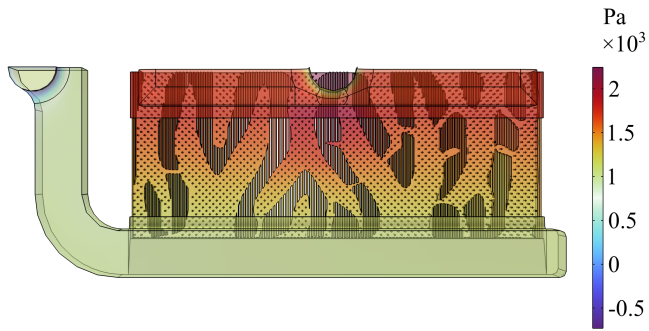
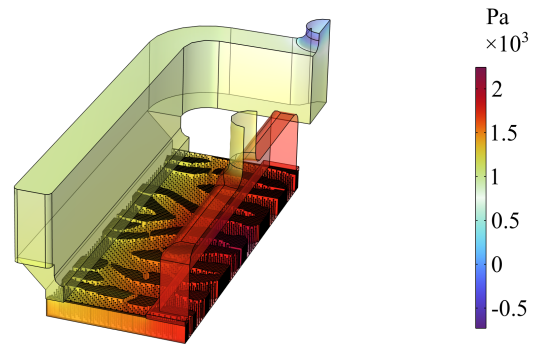


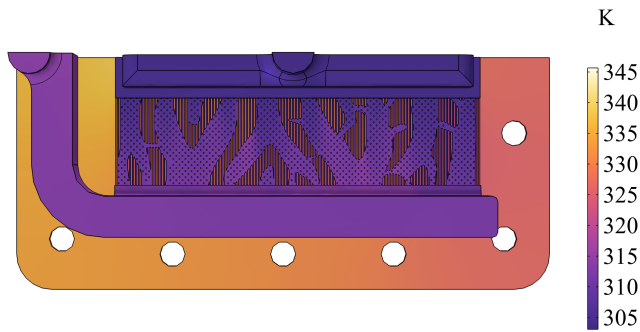
Fig. 7: Final geometry obtained from topology optimization.



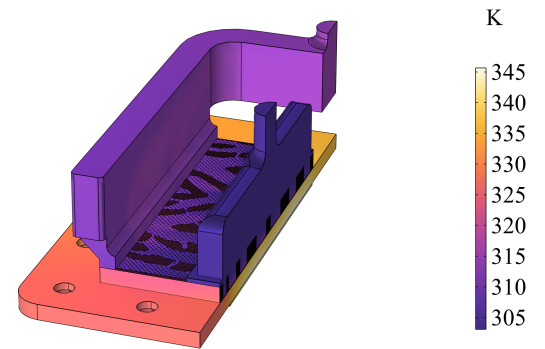
(a) Pressure (view 1).



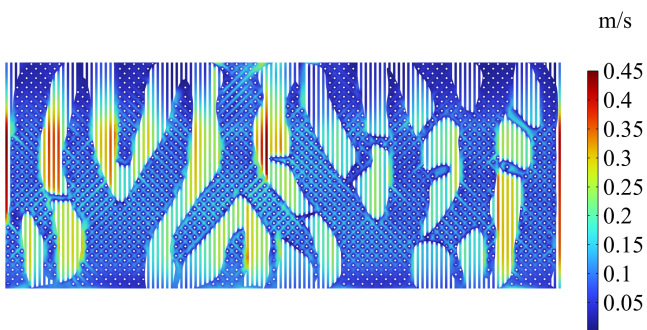
(b) Pressure (view 2).



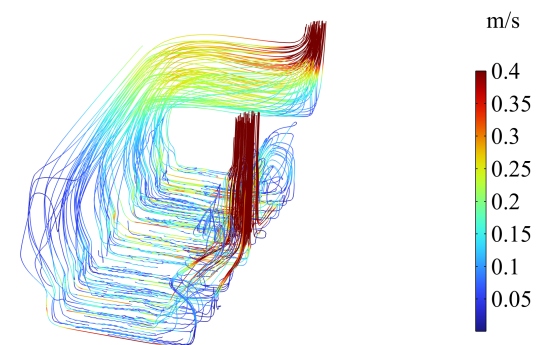
(c) Temperature (view 1).



(d) Temperature (view 2).



(e) Midplane velocity streamlines (view 1).



(f) Velocity (view 2).

Fig. 8: Pressure, temperature, and velocity visualizations for the heat-sink geometry (two views per field).

TABLE I: Figure Of Merit and Model Data

Parameter	Value
Givens	
Total Heater Power, q_{heat} [W] (Zone H = 400 W; Zone BG1 = Zone BG2 = 200 W)	800
Inlet Coolant Temperature, T_{in} [$^{\circ}$ C]	30
Ambient Temperature, T_{amb} [$^{\circ}$ C]	20
Coolant Flow Rate, [$L\ m^{-1}$]	1.2
TTV-ColdPlate TIM Resistance, R_{TIM} [$cm^2\ K\ W^{-1}$]	0.03
Modeled Values — Baseline Design	
Baseline Max Chip Temperature, T_{chip}^{max} [$^{\circ}$ C]	69.68
Baseline Thermal Resistance, $R_{baseline}$ [$K\ W^{-1}$]	0.0496
Pressure Drop, $\Delta p_{baseline}$ [Pa]	1754
Modeled Values — Optimized Design	
Max Chip Temperature, T_{chip}^{max} [$^{\circ}$ C]	72.5
Thermal Resistance, $R_{th,max}$ [$K\ W^{-1}$]	0.0588
Pressure Drop, Δp [Pa]	1200.3
Figure Of Merit (FOM)	0.045

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